

BRAUER GROUPS OF CERTAIN AFFINE CUBIC SURFACES

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ABSTRACT. We study the Brauer groups of affine surfaces that are complements of singular hyperplane sections of smooth cubic surfaces over a field k of characteristic 0. We determine the Brauer group over the algebraic closure as a Galois module for all the possible singular hyperplane sections. For the case when the hyperplane section is geometrically the union of three lines and $k = \mathbb{Q}$, we give examples where the Galois invariant part descends to $\mathbb{Q}$.

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1. INTRODUCTION

The Brauer group of a variety is a geometric object rich with arithmetic information. It is a birational invariant for smooth proper varieties, hence its importance in geometry. In 1970, Manin [11] successfully used the Brauer group to explain failures of the Hasse principle at the time, which sparked arithmetic interest in the method giving birth to the study of Brauer-Manin obstructions on varieties. Naturally, to study applications of the Brauer group one would first attempt to compute it. Determining the Brauer group, and especially its transcendental part, is a difficult problem in general. If the variety is not proper, the problem becomes even more challenging: indeed, one expects the Brauer group to be larger than that of any smooth compactification.

A special class of affine varieties over number fields that attracted arithmetic interest is that of log K3 surfaces. Conjecturally, it is expected that the positivity of the log anticanonical class of an affine variety controls the abundance of integral points: the more positive it is, the more integral points one expects. Log K3 surfaces, having a trivial anticanonical class, are the borderline case, hence their significance. Moreover, as these surfaces are simply connected and can be embedded in smooth proper varieties where the boundary is geometrically a strict normal crossing divisor, computing their Brauer groups is expected to be generally more approachable.

The Brauer groups of different classes of log K3 surfaces over number fields have been studied for arithmetic purposes in the literature. In [5], Colliot-Thélène and

Wittenberg studied the integral Brauer-Manin obstruction on a family of diagonal cubic surfaces, which are complements of smooth genus 1 curves in smooth cubic surfaces. In [2], Bright and Lyczak gave uniform bounds for the size of the Brauer group of complements of a smooth anticanonical divisor in a del Pezzo surface of degree at most 7. In both cases, the Brauer group is related to the arithmetic of the smooth genus 1 anticanonical divisor. A notable example in the singular case is given by Markoff surfaces, where the hyperplane section is the union of three rational lines, was studied by Colliot-Thélène, Wei, and Xu in [4] and by Loughran and Mitankin [9]. In this case, the transcendental Brauer group is always trivial when the ground field is $\mathbb{Q}$.

In this paper, we study the general class of affine surfaces that can be embedded as complements of a singular anticanonical divisor in a smooth cubic surface over a field of characteristic zero. In this situation, the Brauer group is related to the combinatorics of the irreducible (genus 0) components of the boundary divisor and their intersection. We compute the Brauer group over the algebraic closure as a Galois module, which we use to determine all the possibilities of the Brauer group over the ground field. More precisely, we prove the following.

Theorem 1.1. *Let k be a field of characteristic 0 and let $\Gamma_k := \text{Gal}(\bar{k}/k)$. Let X be a smooth cubic surface, H be a hyperplane section, and $U := X \setminus H$. For $d \in k$, let $M_d = (\text{Ind}_{k(\sqrt{d})/k} \mathbb{Z})/\mathbb{Z}$, where $\text{Ind}_{k(\sqrt{d})/k}$ is the induced module from $\Gamma_{k(\sqrt{d})}$ to Γ_k .*

(i) *Suppose that H is the union of a line ℓ and a smooth conic C over k . Then*

$$\text{Br } \bar{U} \cong \begin{cases} 0, & \text{if } \ell \text{ is tangent to } C, \\ \mathbb{Q}/\mathbb{Z}(-1), & \text{if } |\ell(k) \cap C(k)| = 2 \\ \varinjlim (M_d/nM_d(-1)), & \text{if } \ell \cap C = \text{Spec}(k(\sqrt{d})), d \notin k^2. \end{cases}$$

(ii) *Suppose that H is a geometrically irreducible singular cubic curve C . Then*

$$\text{Br } \bar{U} \cong \begin{cases} 0, & \text{if } C \text{ is cuspidal,} \\ \mathbb{Q}/\mathbb{Z}(-1), & \text{if } C \text{ is nodal split multiplicative} \\ \varinjlim (M_d/nM_d(-1)), & \text{if } C \text{ is nodal splitting over } k(\sqrt{d}), d \notin k^2. \end{cases}$$

(iii) *Suppose that $\bar{H}$ is the union of three line ℓ_1, ℓ_2, ℓ_3 . Let L be the minimal extension over which each ℓ_i is defined and let $\tilde{L}$ be its Galois closure. Then*

$$\text{Br } \bar{U} \cong \begin{cases} 0, & \text{if the lines meet at an Eckard point,} \\ \mathbb{Q}/\mathbb{Z}(-1), & \text{if } L = k \text{ and } \cap \ell_i = \emptyset, \text{ or if } \text{Gal}(L/k) \cong \mathbb{Z}/3\mathbb{Z} \\ \varinjlim (M_d/nM_d(-1)), & \text{if } \text{Gal}(\tilde{L}/k) \cong S_3 \text{ or } \text{Gal}(L/k) \cong \mathbb{Z}/2\mathbb{Z}. \end{cases}$$

As opposed to the Markoff surfaces case, the transcendental part is not always trivial over $\mathbb{Q}$. Indeed, all the possible $(\text{Br } \bar{U})^{\Gamma_{\mathbb{Q}}}$ structures in case (iii) of Theorem 1.1 descend to $\mathbb{Q}$.

Theorem 1.2. *Let A be any of the following groups*

$$\mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/4\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

Then there exists an affine cubic surface U over $\mathbb{Q}$ that is the complement of three geometric lines such that $\text{Br}(U)/\text{Br}_1(U) \cong A$.

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2. ALGEBRAIC BRAUER GROUP

Let k is a field of characteristic 0. In this section, we study the algebraic part of the Brauer group of the complement of a singular hyperplane section in a smooth cubic surface.

Lemma 2.1. *Let $X \subset \mathbb{P}^3$ be a smooth cubic surface over field k of characteristic 0 and let H be a hyperplane section. If H is singular, then it is one the following:*

- (i) *a smooth conic and a line over k ;*
- (ii) *a geometrically integral singular cubic curve over k ;*
- (iii) *three geometric lines intersecting at a single (Eckard) point;*
- (iv) *three geometric lines intersecting at three geometric points with a $\text{Gal}(k(C)/k)$ -action where $\text{Gal}(k(C)/k)$ is isomorphic to either 0, $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, or S_3 ;*

Remark 2.2. In Lemma Theorem 2.1, the cases where H is a strict normal crossings divisor are (iv) and (i) when the conic intersects the lines at a degree two point.

Lemma 2.3 ([9, Lemma 2.1]). *Let X be a smooth cubic surface over k , H be a hyperplane section and $U := X \setminus H$. Then $\mathcal{O}(U)^* = k^*$.*

Lemma 2.4. *Let X be a smooth cubic surface over k , H be a hyperplane section and $U := X \setminus H$. Suppose that H is a strict normal crossings divisor. If k is a number field or if $U(k) \neq \emptyset$, then $\text{Pic}(\bar{U})$ is torsion-free and*

$$\text{Br}_1(U)/\text{Br}(k) \cong H^1(k, \text{Pic}(\bar{U})).$$

Moreover, $\text{Br}_1(U)$ is finite and has only finitely many possibilities.

Proof. We first show that $\text{Pic}(\bar{U})$ is torsion free. By Remark 2.2 and Lemma Theorem 2.1, it suffices to consider the cases when the hyperplane section is geometrically the union of three lines, or the union of a line and a smooth conic intersecting at a degree 2 point. The first case follows by [4, Prop 2.2]. Now assume that $X \cap H$ is the union a line L and a smooth conic C both defined over k . Then as $\mathcal{O}(U)^* = k^*$, we have an exact sequence

$$0 \rightarrow \mathbb{Z}[L] \oplus \mathbb{Z}[C] \rightarrow \text{Pic}(\bar{X}) \rightarrow \text{Pic}(\bar{U}) \rightarrow 0.$$

By [8, Proposition 4.8] we have

$$[C] = 3\ell - \sum_{i=1}^6 e_i - [L]$$

and by [8, Theorem 4.9] the class of L is contained in

$$\{\ell, e_1, \dots, e_6\} \cup \{\ell - e_i - e_j : 1 \leq i < j \leq 6\} \cup \{2\ell - \sum_{i \neq j} e_i : 1 \leq j \leq 6\}$$

Thus, for all the possibilities of $[L]$ we deduce that $[C] = c_0\ell + \sum_{i=1}^6 c_i e_i$ where $c_0 \in \{1, 2\}$ and $c_m = -1$ for some $m \in \{1, \dots, 6\}$ (i.e. the coefficient of e_m in the vector corresponding to $[L]$ is 0). To show that $\text{Pic}(\bar{U})$ is torsion free, we need to show that if $[D] \in \text{Pic}(\bar{X})$ satisfies $n[D] \in \mathbb{Z}[L] + \mathbb{Z}[C]$ for some $n > 0$, then $[D] \in \mathbb{Z}[L] + \mathbb{Z}[C]$. Let $[D] = d_0\ell + \sum_{i=1}^6 d_i e_i$ and assume $n[D] = a[L] + b[C]$ for some $a, b \in \mathbb{Z}$. Then we deduce that $nd_m = -b$ and $nd_0 = \alpha a + c_0 b$ for $\alpha \in \{1, 2\}$, where α is the coefficient of ℓ in the vector corresponding to $[L]$. It suffices to show that $n \mid a$. Now, we have $n \mid b$ and $n \mid \alpha a + c_0 b$ so that $n \mid \alpha a$. If $\alpha \neq 2$, then $n \mid \alpha$ and we are done. If $\alpha = 2$, then $[L] \in \{2\ell - \sum_{i \neq j} e_i : 1 \leq j \leq 6\}$ so that $nd_j = -a$ for some $j \neq m$; thus, $n \mid a$ as desired. This shows that $\text{Pic}(\bar{U})$ is torsion free.

Now, by [14, Lemma 6.3.iv] and Lemma Theorem 2.3 we have

$$\text{Br}_1(U)/\text{Br}(k) \cong H^1(k, \text{Pic}(\bar{U})).$$

If K is the minimal extension over which $\text{Pic}(\bar{X})$ splits, then $H^1(K, \text{Pic}(\bar{U})) = 0$ since $\text{Pic}(\bar{X})$ is a torsion-free Γ_K -module. By the inflation-restriction sequence, we obtain

$$H^1(k, \text{Pic}(\bar{U})) \cong H^1(K/k, \text{Pic}(U_K)).$$

Since K was chosen to be minimal, $\text{Gal}(K/k)$ acts on the lattice $\text{Pic}(\bar{U})$ via some subgroup G (well defined up to conjugacy) of an appropriate Weyl group. The result follows by running through the finitely many subgroups G and computing $H^1(G, \text{Pic}(U_K))$, which is finite. $\square$

When the hyperplane section is geometrically the union of three lines, we have the following description of $\text{Pic}(\bar{U})$ as a Γ_k -module.

Proposition 2.5 ([4, Prop 2.2]). *Let $X \subset \mathbb{P}_k^3$ be a smooth projective cubic surface over a field k of characteristic zero. Suppose a plane $\mathbb{P}_k^2 \subset \mathbb{P}_k^3$ cuts on $\bar{X}$ three lines ℓ_1, ℓ_2, ℓ_3 over $\bar{k}$. Let $U \subset X$ be the complement of this plane. Then the natural map $\bar{k}^\times \rightarrow \bar{k}[U]^\times$ is an isomorphism of Galois modules and the natural map*

$$0 \rightarrow \bigoplus_{i=1}^3 \mathbb{Z}[\ell_i] \rightarrow \text{Pic}(\bar{X}) \rightarrow \text{Pic}(\bar{U}) \rightarrow 0$$

is an exact sequence of Galois lattices.

We now compute the possibilities for algebraic Brauer group when the hyperplane section is geometrically the union of three lines.

Proposition 2.6. *Let $X \subset \mathbb{P}_k^3$ be a smooth projective cubic surface over a field k of characteristic zero. Suppose a plane $\mathbb{P}_k^2 \subset \mathbb{P}_k^3$ cuts on $\bar{X}$ three lines ℓ_1, ℓ_2, ℓ_3 over $\bar{k}$ and let $U \subset X$ be the complement of this plane.*

(1) *If $\{\ell_1, \ell_2, \ell_3\}$ is a single Γ_k -orbit, then we have the following possibilities for $\text{Br}_1(U)$ with corresponding $\text{Br}(X)$:*

$\text{Br}_1(U)$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/4$	$\mathbb{Z}/3$	$\mathbb{Z}/3 \times \mathbb{Z}/3$
$\text{Br}(X)$	0	0	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/3$	$\mathbb{Z}/3 \times \mathbb{Z}/3$

(2) *If $\{\ell_1, \ell_2, \ell_3\}$ consists of two Γ_k -orbits, then we have the following possibilities for $\text{Br}_1(U)$ with corresponding $\text{Br}(X)$:*

$\text{Br}_1(U)$	$\mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$(\mathbb{Z}/2)^3$	$(\mathbb{Z}/2)^3$	$\mathbb{Z}/4$	$\mathbb{Z}/2 \times \mathbb{Z}/4$
$\text{Br}(X)$	0	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2 \times \mathbb{Z}/2$

(3) If $\{\ell_1, \ell_2, \ell_3\}$ is three Γ_k -orbit, i.e. each ℓ_i is defined over k , then we have the following possibilities for $\text{Br}_1(U)$ with corresponding $\text{Br}(X)$:

$\text{Br}_1(U)$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^3$	$(\mathbb{Z}/2)^3$	$(\mathbb{Z}/2)^4$	$\mathbb{Z}/4$	$\mathbb{Z}/2 \times \mathbb{Z}/4$
$\text{Br}(X)$	0	0	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$

Proof. First, we note that (3) was computed in [9, Prop 2.5]. We follow the same approach to compute the other cases. By Lemma Theorem 2.4, we have that $\text{Br}_1(U)/\text{Br}(k) \cong H(k, \text{Pic}(\bar{U}))$. We have an exact sequence

$$0 \rightarrow \text{PDiv} \bar{X} \rightarrow \text{Lines} \bar{X} \rightarrow \text{Pic} \bar{X} \rightarrow 0$$

where $\text{Lines} \bar{X}$ is the free abelian group generated by the lines on $\bar{X}$ and $\text{Pdiv} \bar{X}$ is the subgroup of principal divisors. This along with Proposition 2.5 gives a description of $\text{Pic} \bar{U}$ as the quotient of a Γ_k -permutation module by a Γ_k -submodule. We proceed as in [9, Prop 2.5] to compute $H(k, \text{Pic}(\bar{U}))$ using Magma. $\square$

3. THE BRAUER GROUP OVER $\bar{k}$

Let X be a smooth cubic surface over a field k of characteristic 0, let H be a hyperplane section, and set $U := X \setminus H$. In this section we compute the possibilities of $\text{Br}(\bar{U})$ as Γ_k -module for the different cases in Lemma Theorem 2.1. We use this to deduce the possibilities for the transcendental Brauer Group of U .

3.1. Preliminaries. Let Z be a closed subscheme of X , and let $U := X \setminus Z$. For a sheaf $\mathcal{F}$ on $X_{\text{ét}}$, let

$$\Gamma_Z(X, \mathcal{F}) := \ker(\Gamma(X, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F})),$$

the group of sections of $\mathcal{F}$ supported on Z . Let $H_Z^r(X, \mathcal{F})$, the cohomology of $\mathcal{F}$ with support on Z , be the r th derived functor of the functor $\mathcal{G} \mapsto \Gamma_Z(X, \mathcal{G})$.

Theorem 3.1 ([13, Theorem 9.4]). *For a sheaf $\mathcal{F}$ on $X_{\text{ét}}$ and closed $Z \subset X$, there is a long exact sequence*

$$\cdots \rightarrow H_Z^r(X, \mathcal{F}) \rightarrow H^r(X, \mathcal{F}) \rightarrow H^r(U, \mathcal{F}) \rightarrow H_Z^{r+1}(X, \mathcal{F}) \rightarrow \cdots$$

This sequence is functorial in the pairs $(X, X \setminus Z)$ and $\mathcal{F}$.

The following is a consequence of Gabber's absolute purity theorem (see e.g. [3, Theorem 2.3.1]).

Theorem 3.2 ([13, Theorem 16.1]). *For any smooth k -subvariety $Z \subset X$ of codimension c and any locally constant sheaf $\mathcal{F}$ of $\mathbb{Z}/n\mathbb{Z}$ -modules on X , there are canonical isomorphisms*

$$H^{r-2c}(X, \mathcal{F}(-c)) \xrightarrow{\sim} H_Z^r(X, \mathcal{F})$$

for all $r \geq 0$.

By applying Theorem 3.1 for the sheaf μ_n , and by replacing $H_Z^r(X, \mu_n)$ with $H^{r-2c}(X, \mathbb{Z}/n(-c))$ using Theorem 3.2, we obtain the following version of the Gysin sequence (see e.g. [1, (2)]).

Lemma 3.3 (Gysin sequence). *Let X be a smooth variety over a field k of characteristic 0 and $Z \subset X$ a closed reduced smooth subscheme of codimension c . Then there is a long exact sequence*

$$\begin{aligned} 0 \rightarrow H^{2c-1}(X, \mu_n) \rightarrow H^{2c-1}(X \setminus Z, \mu_n) \rightarrow H^0(Z, \mathbb{Z}/n(-c)) \rightarrow \\ \rightarrow H^{2c}(X, \mu_n) \rightarrow H^{2c}(X \setminus Z, \mu_n) \rightarrow H^1(Z, \mathbb{Z}/n(-c)) \rightarrow \dots \end{aligned} \quad (3.1)$$

We also use the following version which allows for Z to be singular (see e.g. [1, Corollary 2.4]).

Lemma 3.4 (Semi-purity in codimension 1). *Let X be a smooth variety over a field k of characteristic 0 and $Z \subset X$ be a reduced, closed subscheme everywhere of codimension 1. Suppose that the singular locus S of Z is of codimension c in Z . Write X° for $X \setminus S$ and Z° for $Z \setminus S$. Then there is a long exact sequence*

$$\begin{aligned} 0 \rightarrow H^1(X, \mu_n) \rightarrow H^1(X \setminus Z, \mu_n) \rightarrow H^0(Z, \mathbb{Z}/n\mathbb{Z}) \rightarrow \\ \rightarrow H^2(X, \mu_n) \rightarrow H^2(X \setminus Z, \mu_n) \rightarrow H^1(Z^\circ, \mathbb{Z}/n\mathbb{Z}) \rightarrow \\ \rightarrow H^3(X^\circ, \mu_n) \rightarrow H^3(X \setminus Z, \mu_n) \dots \end{aligned} \quad (3.2)$$

Lemma 3.5. *Let X be a smooth proper surface over a field k of characteristic 0 and let $U \subset X$ be non-proper subscheme. Then*

- (1) $H^4(\bar{U}, \mathbb{Z}/n(c)) = 0$ for all c .
- (2) If $\text{Pic}(\bar{X})$ is torsion-free, then $H^3(\bar{X}, \mathbb{Z}/n(c)) = 0$ for all c .

Proof. (1) By the Artin comparison theorem (see e.g. [13, Theorem 21.1]), we have

$$H_{\text{ét}}^4(\bar{U}, \mathbb{Z}/n(c)) = H_{\text{sing}}^4(\bar{U}_{\text{cx}}, \mathbb{Z}/n(c))$$

where $\bar{U}_{\text{cx}}$ is $\bar{U}$ is regarded as a complex manifold. Since $\bar{U}_{\text{cx}}$ is not compact, we have $H_{\text{sing}}^4(\bar{U}_{\text{cx}}, \mathbb{Z}/n(c)) = 0$, as desired.

(2) By Poincaré duality (see e.g. [13, Theorem 24.1]), we have

$$H^3(\bar{X}, \mathbb{Z}/n(c)) \cong H^1(\bar{X}, \mu_n)$$

as abelian groups. From the Kummer exact sequence

$$0 \rightarrow \mu_n \rightarrow \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \rightarrow 0,$$

we obtain the exact sequence

$$0 \rightarrow H^1(\bar{X}, \mu_n) \rightarrow \text{Pic}(\bar{X}) \xrightarrow{n} \text{Pic}(\bar{X}).$$

Since $\text{Pic}(\bar{X})$ is torsion-free, we deduce that $H^1(\bar{X}, \mu_n) = 0$ as desired. $\square$

Lemma 3.6. *Let X be a smooth surface over k such that $\text{Pic}(\bar{X})$ is torsion free. Let P be a non-empty codimension 0 finite k -subscheme of X . Then the Gysin sequence induces the exact sequence*

$$0 \rightarrow H^3(\bar{X} \setminus P, \mu_n) \xrightarrow{\psi} H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1)) \rightarrow H^4(\bar{X}, \mu_n) \rightarrow 0.$$

If moreover $P \in X(k)$, then $H^3(\bar{X} \setminus P, \mu_n) = 0$ and the map

$$H^0(P, \mathbb{Z}/n\mathbb{Z}(-1)) \rightarrow H^4(\bar{X}, \mu_n) \quad (3.3)$$

is an isomorphism.

Proof. Applying the Gysin sequence to (X, P) we obtain the exact sequence

$$H^3(\bar{X}, \mu_n) \rightarrow H^3(\bar{X} \setminus P, \mu_n) \xrightarrow{\psi} H^0(P, \mathbb{Z}/n\mathbb{Z}(-1)) \rightarrow H^4(\bar{X}, \mu_n) \rightarrow H^4(\bar{X} \setminus P, \mu_n).$$

By Lemma Theorem 3.5, we have $H^4(\bar{X} \setminus P, \mu_n) = 0$ and $H^3(\bar{X}, \mu_n) = 0$, which proves the first statement. If $P \in X(k)$, then, by Poincaré duality, $H^4(\bar{X}, \mu_n)$ and $H^0(P, \mathbb{Z}/n\mathbb{Z}(-1))$ have the same cardinality, which completes the proof. $\square$

Lemma 3.7. *Let X be a smooth cubic surface over a field k of characteristic 0. Let $\ell \cong \mathbb{P}^1 \subset X$ and let $Q_1, Q_2 \in \ell(k)$. Set $X' := X \setminus \{Q_1, Q_2\}$, $\ell' := \ell \setminus \{Q_1, Q_2\}$, and $U := X \setminus \ell$.*

(i) *There exists a commutative diagram of Γ_k -modules with exact rows*

$$\begin{array}{ccccc} H^3(\bar{X}', \mu_n) & \xhookrightarrow{\psi} & \oplus_{i=1}^2 H^0(\bar{Q}_i, \mathbb{Z}/n\mathbb{Z}(-1)) & \longrightarrow & H^4(\bar{X}, \mu_n) \\ \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ \mathbb{Z}/n\mathbb{Z}(-1) & \xrightarrow{\partial} & \mathbb{Z}/n\mathbb{Z}(-1) \oplus \mathbb{Z}/n\mathbb{Z}(-1) & \xrightarrow{d} & \mathbb{Z}/n\mathbb{Z}(-1) \end{array} \quad (3.4)$$

such that $d(a, b) = a + b$ for $a, b \in \mathbb{Z}/n\mathbb{Z}(-1)$, and $\partial(a) = (a, -a)$ for $a \in \mathbb{Z}/n\mathbb{Z}(-1)$.

(ii) *The map $\varphi : H^1(\bar{\ell}', \mathbb{Z}/n) \rightarrow H^3(\bar{X}', \mu_n)$ induced by the Gysin sequence is an isomorphism producing a commutative diagram*

$$\begin{array}{ccc} H^1(\bar{\ell}', \mathbb{Z}/n) & \xrightarrow{\oplus \partial_{Q_i}} & \oplus_{i=1}^2 H^0(\bar{Q}_i, \mathbb{Z}/n\mathbb{Z}(-1)) \\ & \searrow \varphi & \nearrow \psi \\ & H^3(\bar{X}', \mu_n) & \end{array}$$

where ∂_{Q_i} are the (Gysin) residue maps.

Proof. (i) By applying Lemma Theorem 3.6 to (X, Q) we obtain the exact sequence

$$0 \rightarrow H^3(\bar{X}', \mu_n) \xrightarrow{\psi} H^0(\bar{Q}_1, \mathbb{Z}/n(-1)) \oplus H^0(\bar{Q}_2, \mathbb{Z}/n(-1)) \rightarrow H^4(\bar{X}, \mu_n) \rightarrow 0. \quad (3.5)$$

Applying the Gysin sequence to (X, ℓ') we obtain the exact sequence

$$H^2(\bar{X}, \mu_n) \rightarrow H^2(\bar{U}, \mu_n) \rightarrow H^1(\bar{\ell}', \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\varphi} H^3(\bar{X}', \mu_n) \rightarrow H^3(\bar{U}, \mu_n), \quad (3.6)$$

and to (X, ℓ) we get the exact sequence

$$H^2(\bar{X}, \mu_n) \rightarrow H^2(\bar{U}, \mu_n) \rightarrow H^1(\bar{\ell}, \mathbb{Z}/n\mathbb{Z}) \cong 0$$

By (3.6), this gives an exact sequence

$$0 \rightarrow H^1(\bar{\ell}', \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\varphi} H^3(\bar{X}', \mu_n) \rightarrow H^3(\bar{U}, \mu_n).$$

From (3.5) we see that $H^3(\bar{X}', \mu_n)$ has cardinality n which is equal to that of $H^1(\bar{\ell}', \mu_n)$. Therefore, we obtain an isomorphism

$$\varphi : H^1(\bar{\ell}', \mathbb{Z}/n) \xrightarrow{\sim} H^3(\bar{X}', \mu_n).$$

Now, the vertical maps in (3.4) come from the maps

$$H^1(\bar{\ell}', \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} H^1(\mathbb{G}_m, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}(-1),$$

$H^0(\bar{Q}_i, \mathbb{Z}/n\mathbb{Z}(-1)) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}(-1)$, and $H^4(\bar{X}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}(-1)$ (where the last one follows by Poincaré duality). Then there exists maps ∂ and d producing the commutative diagram (3.4). It remains to check that d is the claimed map.

By purity (Theorem 3.2), the map d is induced by the map

$$H_{\bar{Q}_1}^4(\bar{X}, \mu_n) \oplus H_{\bar{Q}_2}^4(\bar{X}, \mu_n) \xrightarrow{d} H^4(\bar{X}, \mu_n).$$

Given an injective resolution of μ_n ,

$$0 \rightarrow \mu_n \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1 \rightarrow \mathcal{I}^2 \rightarrow \cdots,$$

the map d is then induced by the map

$$\Gamma_{Q_1}(X, \mathcal{I}^4) \oplus \Gamma_{Q_2}(X, \mathcal{I}^4) \rightarrow \Gamma(X, \mathcal{I}^4),$$

which takes a section s_1 of $\mathcal{I}^4$ supported on Q_1 and a section s_2 supported on Q_2 to their sum $s_1 + s_2$ in $\Gamma(X, \mathcal{I}^4)$. This proves that d is the claimed map. Note that by the exactness of the bottom row in (3.4), ∂ has to be the claimed map as well.

(ii) By applying the Gysin sequence to (ℓ, ℓ') and by using that $H^1(\mathbb{P}^1, \mathbb{Z}/n) \cong 0$ and $H^2(\mathbb{P}^1, \mathbb{Z}/n) \cong \mathbb{Z}/n(-1)$, we obtain the exact sequence

$$0 \rightarrow H^1(\bar{\ell}', \mathbb{Z}/n) \xrightarrow{\oplus \partial_{Q_i}} \oplus_{i=1}^2 H^0(\bar{Q}_i, \mathbb{Z}/n\mathbb{Z}(-1)) \rightarrow \mathbb{Z}/n(-1)$$

where the last map is the sum of the residues. The result now follows by noting that φ is induced by the map

$$\Gamma_{\ell'}(X, \mathcal{I}^3) \hookrightarrow \Gamma(X, \mathcal{I}^3).$$

□

3.2. Proof of Theorem 1.1. First, from the Kummer exact sequence

$$0 \rightarrow \mu_n \rightarrow \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \rightarrow 0,$$

we obtain the following commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\text{Pic } \bar{X})/n & \longrightarrow & H^2(\bar{X}, \mu_n) & \longrightarrow & (\text{Br } \bar{X})[n] \longrightarrow 0 \\ & & \downarrow & & \downarrow \Psi_n & & \downarrow \\ 0 & \longrightarrow & (\text{Pic } \bar{U})/n & \longrightarrow & H^2(\bar{U}, \mu_n) & \longrightarrow & (\text{Br } \bar{U})[n] \longrightarrow 0 \end{array} \quad (3.7)$$

Lemma 3.8. *We have an isomorphism*

$$\text{Br}(\bar{U})[n] \cong \text{coker}(\Psi_n).$$

Proof. The left vertical map in (3.7) is surjective as X is smooth. Since X is a geometrically rational surface, we have $\text{Br}(\bar{U}) = 0$. The result now follows by the Snake lemma. □

In light of Lemma Theorem 3.8, it suffices to compute $\text{coker}(\Psi_n)$, for all n , for all the different possible singular hyperplane sections.

Case 1: a smooth conic and a line both defined over k

Proposition 3.9. *Suppose that H is the union of a line ℓ and a smooth conic C both defined over k . If ℓ meets C at rational points, then as Γ_k -modules we have*

$$\text{Br } \bar{U} \cong \begin{cases} 0, & \text{if } \ell \text{ is tangent to } C, \\ \mathbb{Q}/\mathbb{Z}(-1) := \text{Hom}(\mu_\infty, \mathbb{Q}/\mathbb{Z}), & \text{if } |\ell(k) \cap C(k)| = 2 \end{cases}$$

If ℓ meets C at a degree 2 point with residue field $L = k(\sqrt{d})$, $d \notin k^2$, then as Γ_k -modules

$$(\text{Br } \bar{U})[n] \cong M_d/nM_d(-1),$$

for all $n \in \mathbb{Z}_{>0}$, where $M_d = (\text{Ind}_{k(\sqrt{d})/k} \mathbb{Z})/\mathbb{Z}$.

Proof. Let $U_1 = X \setminus C$. Applying the Gysin sequence to (X, U_1) we get the exact sequence

$$H^2(\bar{X}, \mu_n) \rightarrow H^2(\bar{U}_1, \mu_n) \rightarrow H^1(\bar{C}, \mathbb{Z}/n\mathbb{Z}) \rightarrow \cdots \quad (3.8)$$

Since $\bar{C} \cong \mathbb{P}^1$ we have $H^1(\bar{C}, \mathbb{Z}/n\mathbb{Z}) = 0$ so $H^2(\bar{X}, \mu_n) \rightarrow H^2(\bar{U}_1, \mu_n)$ is surjective. The exact sequence (3.8) continues to a long exact sequence

$$H^3(\bar{X}, \mu_n) \rightarrow H^3(\bar{U}_1, \mu_n) \rightarrow H^2(\bar{C}, \mathbb{Z}/n\mathbb{Z}) \rightarrow H^4(\bar{X}, \mu_n) \rightarrow H^4(\bar{U}_1, \mu_n).$$

By Lemma Theorem 3.5, we have $H^4(\bar{U}_1, \mu_n) = 0$ and $H^3(\bar{X}, \mu_n) = 0$. By Poincaré duality, $H^2(\bar{C}, \mathbb{Z}/n\mathbb{Z})$ and $H^4(\bar{X}, \mu_n)$ are isomorphic to $\mathbb{Z}/n$ as abelian groups, which implies that $H^2(\bar{C}, \mathbb{Z}/n\mathbb{Z}) \rightarrow H^4(\bar{X}, \mu_n)$ is an isomorphism and that $H^3(\bar{U}_1, \mu_n) = 0$. Applying the Gysin sequence to $(U_1, U_1 \cap \ell)$ we get

$$H^2(\bar{U}_1, \mu_n) \rightarrow H^2(\bar{U}, \mu_n) \rightarrow H^1(\bar{U}_1 \cap \bar{\ell}, \mathbb{Z}/n\mathbb{Z}) \rightarrow H^3(\bar{U}_1, \mu_n) = 0.$$

Since $H^2(\bar{X}, \mu_n) \rightarrow H^2(\bar{U}_1, \mu_n)$ is surjective, we have

$$(\text{Br } \bar{U})[n] \cong H^1(\bar{U}_1 \cap \bar{\ell}, \mathbb{Z}/n\mathbb{Z}).$$

If ℓ is tangent to C , then $U_1 \cap \ell \cong \mathbb{A}^1$, so that $H^1(\bar{U}_1 \cap \bar{\ell}, \mathbb{Z}/n\mathbb{Z}) = 0$. If ℓ intersects C in two distinct k -points, then $U_1 \cap \ell \cong \mathbb{G}_m$. By Kummer theory, $H^1(\mathbb{G}_{m, \bar{k}}, \mu_n) = \mathbb{Z}/n\mathbb{Z}$ so that $H^1(\mathbb{G}_{m, \bar{k}}, \mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}(-1)$.

If ℓ meets C at a degree 2 point with residue field $L = k(\sqrt{d})$, then

$$\ell \setminus P \cong \text{Res}_{L/K} \mathbb{G}_{m, L} / \mathbb{G}_{m, k} =: T,$$

is a 1-dimensional torus over k , where $\text{Res}_{L/K}$ is the Weil restriction. Let $G := \text{Gal}(L/k) \cong \mathbb{Z}/2\mathbb{Z}$. We have an exact sequence

$$1 \rightarrow \bar{k}^\times \rightarrow H^0(\bar{T}, \mathbb{G}_m) \rightarrow U(\bar{T}) \rightarrow 1, \quad (3.9)$$

where $U(\bar{T}) = \text{Hom}_{k\text{-groups}}(\bar{T}, \mathbb{G}_{m, \bar{k}})$ which is isomorphic to $\mathbb{Z}$ as an abelian group. Note that the action of G on $U(\bar{T})$ translates to $\mathbb{Z}$ as the action induced by $G \rightarrow \text{Aut}(\mathbb{Z}) \cong \{\pm 1\}$. More concretely, we have

$$U(\bar{T}) \cong (\text{Ind}_{L/k} \mathbb{Z})/\mathbb{Z} =: M$$

as Γ_k -modules, where the $\mathbb{Z} \hookrightarrow \text{Ind}_{L/k} \mathbb{Z}$ is the diagonal embedding. Since T has a rational point, we see that (3.9) splits, i.e.

$$H^0(\bar{T}, \mathbb{G}_m) \cong \bar{k}^\times \oplus U(\bar{T}).$$

Hence, from the Kummer sequence, and using that $H^1(\bar{T}, \mathbb{G}_m) = 0$, we deduce that $H^1(\bar{T}, \mu_n) \cong M_d/nM_d$. Therefore, $(\text{Br } \bar{U})[n] \cong H^1(\bar{T}, \mathbb{Z}/n\mathbb{Z}) \cong M_d/nM_d(-1)$. $\square$

Case 2: a geometrically irreducible singular cubic curve over k

Proposition 3.10. *Suppose that H is a geometrically irreducible singular cubic curve C . Then as Γ_k -modules we have*

$$\text{Br } \bar{U} \cong \begin{cases} 0, & \text{if } C \text{ is cuspidal,} \\ \mathbb{Q}/\mathbb{Z}(-1) := \text{Hom}(\mu_\infty, \mathbb{Q}/\mathbb{Z}), & \text{if } C \text{ is nodal split multiplicative} \end{cases}$$

If C is nodal non-split multiplicative splitting over $L = k(\sqrt{d})$, $d \notin k^2$, then

$$(\text{Br } \bar{U})[n] \cong M_d/nM_d(-1)$$

as Γ_k -modules for all n , where $M_d := (\text{Ind}_{k(\sqrt{d})/k} \mathbb{Z})/\mathbb{Z}$.

Proof. First, note that the singular point P of C has to be a k -point. Set $X' = X \setminus \{P\}$ and $C' = C \setminus \{P\}$. Since P is of codimension 2 in X , by purity, we may replace X with X' in diagram (3.7). By Lemma Theorem 3.6, we have that $H^3(\bar{X}', \mu_n) = 0$. Thus, by applying the Gysin sequence to (X', C') we get the exact sequence

$$H^2(\bar{X}', \mu_n) \rightarrow H^2(\bar{U}, \mu_n) \rightarrow H^1(\bar{C}', \mathbb{Z}/n\mathbb{Z}) \rightarrow 0.$$

If C is cuspidal, then $H^1(C'_k, \mathbb{Z}/n\mathbb{Z}) = H^1(\mathbb{A}^1, \mathbb{Z}/n\mathbb{Z}) = 0$. If C is nodal split multiplicative, then $H^1(C'_k, \mathbb{Z}/n\mathbb{Z}) = H^1(\mathbb{G}_{m,\bar{k}}, \mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}(-1)$. Finally, if C is nodal non-split multiplicative, then $C' \cong \text{Res}_{L/k} \mathbb{G}_{m,F}/\mathbb{G}_{m,k}$ so that $H^1(C'_k, \mathbb{Z}/n\mathbb{Z}) \cong M_d/nM_d(-1)$. $\square$

Case 3: three geometric lines with a Galois action

Suppose that $C := X \cap H$ is geometrically the union of three lines and let L be the minimal extension over which they are defined. We assume that the lines do not intersect in an Eckard point, as in this case $\text{Br } \bar{U} = 0$ (see [9, Prop 2.4]). Let P be the singular locus of C and set $X' := X \setminus P$, $C' := C \setminus P$. We can write $P_L = \cup_{i=1}^3 P_i$, where each $P_i \in X_L(L)$, and $C_L := \cup_{i=1}^3 \ell_i$, where each $\ell_i \cong \mathbb{P}_L^1$ and such that $\cup_{j \neq i} P_j \subset \ell_i$. Set $\ell'_i = \ell_i \setminus \cup_{j \neq i} P_j$ so that $C'_L = \cup_{i=0}^3 \ell'_i$. We list all the possibilities for P :

- (i) $P = \sqcup_{i=1}^3 \text{Spec}(k)$. In this case, $\text{Br } \bar{U} = 0$ (see [9, Prop 2.4]).
- (ii) $P = \text{Spec}(L) \sqcup \text{Spec}(k)$ where $L := k(\sqrt{d})$, $d \notin k^2$ and $\text{Gal}(L/k) \cong \mathbb{Z}/2\mathbb{Z}$.
- (iii) $P = \text{Spec}(L)$ where L is a cubic Galois extension of k and $\text{Gal}(L/k) \cong \mathbb{Z}/3\mathbb{Z}$.
- (iv) $P = \text{Spec}(L)$ where L is a cubic extension of k with Galois closure $\tilde{L} = L(\sqrt{d})$ for some $d \notin k^2$. In this case, $\text{Gal}(\tilde{L}/k) \cong S_3$.

Proposition 3.11. *As Γ_k -modules, we have*

$$\mathrm{Br}(\bar{U}) \cong \begin{cases} \mathbb{Q}/\mathbb{Z}(-1), & \text{if } L = k \text{ or if } \mathrm{Gal}(L/k) \cong \mathbb{Z}/3\mathbb{Z} \\ \varinjlim (M_d/nM_d(-1)), & \text{if } \mathrm{Gal}(\bar{L}/k) \cong S_3 \text{ or } \mathrm{Gal}(L/k) \cong \mathbb{Z}/2\mathbb{Z} \end{cases}$$

where $M_d := (\mathrm{Ind}_{k(\sqrt{d})/k} \mathbb{Z})/\mathbb{Z}$.

Proof. By Lemma Theorem 3.6, we have an exact sequence of Γ_k -modules

$$0 \rightarrow H^3(\bar{X}', \mu_n) \xrightarrow{\psi} H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1)) \rightarrow H^4(\bar{X}, \mu_n) \rightarrow 0 \quad (3.10)$$

Applying the Gysin sequence to (X', C') we obtain the exact sequence of Γ_k -modules

$$H^2(\bar{X}', \mu_n) \rightarrow H^2(\bar{U}, \mu_n) \rightarrow H^1(\bar{C}', \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\varphi} H^3(\bar{X}', \mu_n) \rightarrow H^3(\bar{U}, \mu_n). \quad (3.11)$$

We wish to compute $\ker(\varphi)$ as a Γ_k -module. We first compute it as a Γ_L -module. Note that $H^1(\bar{C}', \mathbb{Z}/n\mathbb{Z}) = \bigoplus_{i=1}^3 H^1(\bar{\ell}'_i, \mathbb{Z}/n\mathbb{Z})$ and $H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1)) = \bigoplus_{i=1}^3 H^0(\bar{P}_i, \mathbb{Z}/n\mathbb{Z}(-1))$ as Γ_L -modules.

Step 1: explicit description of the map φ .

For all $j \in \{1, 2, 3\}$, by applying Theorem 3.1 to the pairs $(X, X \setminus P)$ and $(X, X \setminus \cup_{i \neq j} P_i)$, we obtain the following commutative diagram with exact rows:

$$\begin{array}{ccccc} H^3(\bar{X}', \mu_n) & \xrightarrow{\psi} & \bigoplus_{i=1}^3 H^4_{P_i}(\bar{X}, \mu_n) & \longrightarrow & H^4(\bar{X}, \mu_n) \\ \uparrow & & \uparrow & & \uparrow \\ H^3(\bar{X} \setminus \cup_{i \neq j} P_i, \mu_n) & \longrightarrow & \bigoplus_{i \neq j} H^4_{P_i}(\bar{X}, \mu_n) & \longrightarrow & H^4(\bar{X}, \mu_n) \end{array} \quad (3.12)$$

By applying Theorem 3.1 to the pairs $(X', X \setminus \ell'_j)$ and $(X \setminus \cup_{i \neq j} P_i, X \setminus \ell_i)$ for each j , we obtain a commutative diagram

$$\begin{array}{ccc} H^3_{\ell'_j}(\bar{X}', \mu_n) & \xrightarrow{\varphi_j} & H^3(\bar{X}', \mu_n) \\ \uparrow & & \uparrow \\ H^3_{\ell'_j}(\bar{X} \setminus \cup_{i \neq j} P_i, \mu_n) & \longrightarrow & H^3(\bar{X} \setminus \cup_{i \neq j} P_i, \mu_n) \end{array} \quad (3.13)$$

so that $\bigoplus_{j=1}^3 \varphi_j$ is the map $\varphi : \bigoplus_{i=1}^3 H^1(\bar{\ell}'_i, \mathbb{Z}/n\mathbb{Z}) \rightarrow H^3(\bar{X}', \mu_n)$ in (3.11).

By purity (Theorem 3.2) and by combining the diagrams (3.12) and (3.13), we obtain the following commutative diagram

$$\begin{array}{ccccc} H^1(\bar{\ell}'_j, \mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\psi \circ \varphi_j} & \bigoplus_{i=1}^3 H^4_{P_i}(\bar{X}, \mu_n) & \longrightarrow & H^4(\bar{X}, \mu_n) \\ \wr \uparrow & & \uparrow & & \uparrow \\ H^1(\bar{\ell}'_j, \mathbb{Z}/n\mathbb{Z}) & \longrightarrow & \bigoplus_{i \neq j} H^4_{P_i}(\bar{X}, \mu_n) & \longrightarrow & H^4(\bar{X}, \mu_n) \end{array}$$

for all $j \in \{1, 2, 3\}$.

Applying Lemma Theorem 3.7 to ℓ_j , for each j , we obtain a commutative diagram of Γ_L -modules

$$\begin{array}{ccccc} H^1(\bar{\ell}'_j, \mathbb{Z}/n) & \xrightarrow{\psi \circ \varphi_j} & \oplus_{i \neq j} H^0(P_i, \mathbb{Z}/n(-1)) & \longrightarrow & H^4(\bar{X}, \mu_n) \\ \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ \mathbb{Z}/n\mathbb{Z}(-1) & \xrightarrow{\partial_j} & \oplus_{i \neq j} \mathbb{Z}/n\mathbb{Z}(-1) & \xrightarrow{d_j} & \mathbb{Z}/n\mathbb{Z}(-1) \end{array}$$

such that $d_j(a, b) = a + b$ for $a, b \in \mathbb{Z}/n\mathbb{Z}(-1)$ and $\partial_j(a) = (a, -a)$ for $a \in \mathbb{Z}/n\mathbb{Z}(-1)$. Therefore, by combining these diagrams for all j , we obtain a commutative diagram of Γ_L -modules

$$\begin{array}{ccc} \oplus_{i=1}^3 H^1(\bar{\ell}'_i, \mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\psi \circ \varphi} & \oplus_{i=1}^3 H^0(\bar{P}_i, \mathbb{Z}/n\mathbb{Z}(-1)) \\ \downarrow \wr & & \downarrow \wr \\ \oplus_{i=1}^3 \mathbb{Z}/n\mathbb{Z}(-1) & \xrightarrow{\partial = \sum_{i=1}^3 \partial_i} & \oplus_{i=1}^3 \mathbb{Z}/n\mathbb{Z}(-1) \end{array} \quad (3.14)$$

such that the restriction of $\psi \circ \varphi$ to $H^1(\bar{\ell}'_i, \mathbb{Z}/n)$ is $\psi \circ \varphi_i$, and

$$\partial(a, b, c) = (c - b, a - c, b - a) \quad (3.15)$$

for $(a, b, c) \in \oplus_{i=1}^3 \mathbb{Z}/n\mathbb{Z}(-1)$. Thus, since ψ is injective, we have

$$\ker(\varphi) \cong \ker(\partial) = \{(a, a, a) \in \oplus_{i=1}^3 \mathbb{Z}/n\mathbb{Z}(-1)\} \cong \mathbb{Z}/n\mathbb{Z}(-1) \quad (3.16)$$

as Γ_L -modules.

Step 2: $\ker(\varphi)$ as a Γ_k -module.

So far, by step 1, we have the following commutative diagram of abelian groups

$$\begin{array}{ccc} H^1(\bar{C}', \mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\psi \circ \varphi} & H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1)) \\ \downarrow \wr & & \downarrow \wr \\ \oplus_{i=1}^3 \mathbb{Z}/n\mathbb{Z} & \xrightarrow{\partial = \sum_{i=1}^3 \partial_i} & \oplus_{i=1}^3 \mathbb{Z}/n\mathbb{Z} \end{array} \quad (3.17)$$

where the top morphism is that of Γ_k -modules. Note that

$$H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1)) \cong \text{Ind}_L^k \mathbb{Z}/n\mathbb{Z}(-1)$$

as a Γ_k -module, where Ind_L^k is the induced module from Γ_L to Γ_k . In other words, we have

$$H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1)) \cong \bigoplus_{i=1}^3 (\mathbb{Z}/n\mathbb{Z}(-1))_{P_i}$$

where $g \in \Gamma_k$ acts by permuting the coordinates according to their action on the associated point, followed by its action on each coordinate. We now determine the Γ_k -action on $H^1(\bar{C}', \mathbb{Z}/n\mathbb{Z})$ by using the explicit description of the map ∂ along with the description of the Γ_k -action on $H^0(\bar{P}, \mathbb{Z}/n\mathbb{Z}(-1))$.

Observe that the Γ_k -action on the points $\{\bar{P}_i\}_{i=1}^3$ determines that on the lines $\{\bar{\ell}'_i\}_{i=1}^3$. Let $\tilde{L}$ be the Galois closure of L in $\bar{k}$ and let $G := \text{Gal}(\tilde{L}/k)$ which is

isomorphic to a subgroup of S_3 via the action on the indices of the points P_i (and hence on the indices of the lines ℓ_i).

- (1) Suppose that $g \in \Gamma_k$ has class in $G = \Gamma_k/\Gamma_L$ equal to the 3-cycle (123). Then by (3.15) we have

$$\partial(g \cdot (a, 0, 0)) = g \cdot \partial(a, 0, 0) = g \cdot (0, a, -a) = (-a^g, 0, a^g) = \partial(0, a^g, 0)$$

where a^g is the action of $g \in \Gamma_k$ on $a \in \mathbb{Z}/n(-1)$. This shows that

$$g \cdot (a, 0, 0) = (0, a^g, 0) + (b, b, b)$$

for some $(b, b, b) \in \ker(\varphi)$. However, since G acts on $\{\ell_i\}_{i=1}^3$ by permutation, we have to have $b = 0$. Similarly, we compute $g \cdot (0, a, 0) = (0, 0, a^g)$ and $g \cdot (0, 0, a) = (a^g, 0, 0)$, which implies that

$$g \cdot (a, a, a) = (a^g, a^g, a^g).$$

If g has class equal to the 3-cycle (132) instead, then a similar computation shows that $g \cdot (a, a, a) = (a^g, a^g, a^g)$ as well.

- (2) Suppose that $g \in \Gamma_k$ has class in $G = \Gamma_k/\Gamma_L$ equal to the 2-cycle (12) (so that g flips the line ℓ_3 and switches the lines ℓ_1 and ℓ_2). Then

$$\partial(g \cdot (a, 0, 0)) = g \cdot \partial(a, 0, 0) = g \cdot (0, a, -a) = (a^g, 0, -a^g) = \partial(0, -a^g, 0)$$

so that $g \cdot (a, 0, 0) = (0, -a^g, 0)$. Similarly, we compute $g \cdot (0, a, 0) = (-a^g, 0, 0)$ and $g \cdot (0, 0, a) = (0, 0, -a^g)$. Hence, we deduce that

$$g \cdot (a, a, a) = (-a^g, -a^g, -a^g).$$

If g has class equal to the 2-cycle (23) or (13), then a similar computation shows that $g \cdot (a, a, a) = (-a^g, -a^g, -a^g)$ as well.

Therefore, we conclude the following:

- If L is Galois and $\text{Gal}(L/k) \cong \mathbb{Z}/3$, then by (1) we deduce that $\ker(\varphi) \cong \mathbb{Z}/n(-1)$ as a Γ_k -module.
- If L is Galois and $\text{Gal}(L/k) \cong \mathbb{Z}/2$, then by (2) we deduce that $\ker(\varphi) \cong M_d/nM_d(-1)$ as a Γ_k -module, where $M := (\text{Ind}_{k(\sqrt{d})/k} \mathbb{Z})/\mathbb{Z}$. Indeed, this follows since $g \in \Gamma_k$ has class in G equal to a 2-cocycle if and only if it conjugate $\sqrt{d}$.
- If $\text{Gal}(\tilde{L}/k) \cong S_3$, then by (1) and (2) we deduce that $\ker(\varphi) \cong M_d/nM_d(-1)$ as a Γ_k -module.

The result now follows by taking the direct limit. $\square$

4. THE TRANSCENDENTAL BRAUER GROUP

In this section, we list all the possibilities for the transcendental Brauer group $\text{Br}(U)/\text{Br}_1(U)$. We do so by computing $(\text{Br}(\bar{U}))^{\Gamma_k}$ for all the different Γ_k -module structures on $\text{Br}(\bar{X})$ listed in Theorem 1.1.

We start with the following lemma.

Lemma 4.1. *Let n be a prime power and $d \in k \setminus k^2$. Set $M_d := (\text{Ind}_{k(\sqrt{d})/k} \mathbb{Z})/\mathbb{Z}$.*

If $\sqrt{d} \notin k(\zeta_n)$, then

$$(M_d/nM_d(-1))^{\Gamma_k} \cong \begin{cases} \mathbb{Z}/2\mathbb{Z}, & \text{if } n = 2^i, i > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If $\sqrt{d} \in k(\zeta_n)$ and $[k(\zeta_n) : k] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$, then

$$(M_d/nM_d(-1))^{\Gamma_k} \cong \begin{cases} \mathbb{Z}/4\mathbb{Z}, & \text{if } n = 2^i, i > 0, \\ \mathbb{Z}/3\mathbb{Z}, & \text{if } n = 3^i, i > 0, \\ 0, & \text{otherwise} \end{cases}$$

Proof. First, note that the action of Γ_k on $M_d/nM_d(-1)$ factors through $\text{Gal}(\mathbb{Q}(\zeta_n, \sqrt{d})/\mathbb{Q})$, which we denote by G_n , where ζ_n is some fixed primitive n th root of unity. We denote $\phi \in \mathbb{Z}/n(-1) = \text{Hom}(\mu_n, \mathbb{Z}/n)$ that maps ζ_n to a by ϕ_a . Let H_n be the G_n -module with underlying group $\text{Hom}(\mu_n, \mathbb{Z}/n)$ equipped with the following G_n -action:

$$\begin{cases} g \cdot \phi \mapsto \{x \mapsto -\phi(g^{-1}x)\}, & \text{if } g(\sqrt{d}) = -\sqrt{d}, \\ g \cdot \phi \mapsto \{x \mapsto \phi(g^{-1}x)\}, & \text{if } g(\sqrt{d}) = \sqrt{d}, \end{cases}$$

for $\phi \in \text{Hom}(\mu_n, \mathbb{Z}/n)$. Then it is easy to see that we have an isomorphism of G_n -modules

$$M_d/nM_d \otimes \text{Hom}(\mu_n, \mathbb{Z}/n) \xrightarrow{\sim} H_n.$$

Suppose that $\sqrt{d} \notin k(\zeta_n)$. Then $G_n = \text{Gal}(\mathbb{Q}(\zeta_n, \sqrt{d})/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times \times \mathbb{Z}/2\mathbb{Z}$. If $g = (0, 1) \in (\mathbb{Z}/n\mathbb{Z})^\times \times \mathbb{Z}/2\mathbb{Z}$, then $g \cdot \phi_a = \phi_{-a}$; hence, if g fixes ϕ_a , then $\phi_{2a} = 0 \in H_n$ i.e. ϕ_a is 2-torsion. This shows that $H_n^{G_n} = \mathbb{Z}/2\mathbb{Z}$ for $n = 2^i, i \geq 1$, and $H_n^{G_n} = 0$ for $n = p^i, p > 2, i \geq 1$.

Suppose now that $\sqrt{d} \in k(\zeta_n)$ and $[k(\zeta_n) : k] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$. Then $G_n = \text{Gal}(k(\zeta_n)/k) \cong (\mathbb{Z}/n\mathbb{Z})^\times$. For $g \in G_n$, we have that $g^{-1}\zeta_n = \zeta_n^t$ for some $t \in \mathbb{Z}$ coprime to n ; we denote g by g_t .

- If $n = 2^i$ and $i > 1$, then $g_3 \in G_n$. If $g_3 \in G_n$ fixes $\phi_m \in H_n$, then $\phi_m = g_3 \cdot \phi_m = \phi_{3m}$ or $\phi_m = g_3 \cdot \phi_m = \phi_{-3m}$, implying that either $2m = 0$ or $4m = 0$ in $\mathbb{Z}/n$. This shows that $H_n^{G_n} \subset \mathbb{Z}/4\mathbb{Z}$. Since $[k(\zeta_n) : k] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$, we deduce that $d \equiv -1 \in k^\times/k^{\times 2}$. Thus, if $\phi_m \in H_4 \setminus H_2$, then $\phi_m(\zeta_n) = i$ or $\phi_m(\zeta_n) = -i$. Therefore, if $g \in G_n$ acts non-trivially on ϕ_m then g is complex conjugation, i.e. $g(\sqrt{d}) = -\sqrt{d}$; thus,

$$(g \cdot \phi_m)(\zeta_n) = -\phi_m(g^{-1}(\zeta_n)) = -g(i) = i = \phi_m(\zeta_n).$$

This shows that $H_n^{G_n} \cong \mathbb{Z}/4\mathbb{Z}$.

- If $n = 3^i$ and $i > 0$, then $g_2 \in G_n$. If $g_2 \in G_n$ fixes $\phi_m \in H_n$, then $\phi_m = g_2 \cdot \phi_m = \phi_{2m}$ or $\phi_m = g_2 \cdot \phi_m = \phi_{-2m}$, implying that either $m = 0$ or $3m = 0$. This shows that $H_n^{G_n} \subset \mathbb{Z}/3\mathbb{Z}$. Since $[k(\zeta_n) : k] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$, we deduce that $d \equiv -3 \in k^\times/k^{\times 2}$. Thus, if $\phi_m \in H_3 \setminus \{0\}$ then $\phi_m(\zeta_n) = (-1 + \sqrt{-3})/2$ or $\phi_m(\zeta_n) = (-1 - \sqrt{-3})/2$. Therefore, if $g \in G_n$ acts non-trivially on ϕ_m then g conjugates $\sqrt{-3}$; thus,

$$(g \cdot \phi_m)(\zeta_n) = -\phi_m(g^{-1}(\zeta_n)) = g(\zeta_3^{-1}) = \zeta_3 = \phi_m(\zeta_n).$$

This shows that $H_n^{G_n} \cong \mathbb{Z}/3\mathbb{Z}$.

- If $n = p^i$, $i > 0$, and p is a prime ≥ 5 , then $g_2 \in G_n$. Thus, if $g_2 \in G_n$ fixes $\phi_m \in H_n$, then $\phi_m = g_2 \cdot \phi_m = \phi_{2m}$ or $\phi_m = g_2 \cdot \phi_m = \phi_{-2m}$, implying that either $m = 0$ or $3m = 0$ in $\mathbb{Z}/n\mathbb{Z}$. This implies that $m = 0$ as n is a power of a prime greater than 3. This shows that $H_n^{G_n} = 0$.

The result now follows. $\square$

Theorem 4.2. *Let X be a smooth cubic surface over a number field k , H be a hyperplane intersecting S , and $U = X \setminus H$. Assume that the only non-trivial root of unity contained in k is -1 .*

- (1) *Suppose that $H \cap X$ is the union of a line L and a smooth conic C both defined over k . If L intersects C at a k -point, then $\text{Br } U / \text{Br}_1 U$ is a subgroup of*

$$\begin{cases} 0, & \text{if } L \text{ is tangent to } C, \\ \mathbb{Z}/2, & \text{if } |L(k) \cap C(k)| = 2 \end{cases}$$

If L meets C at a degree 2 point with residue field $L = k(\sqrt{d})$, $d \notin k^2$, then $\text{Br } U / \text{Br}_1 U$ is a subgroup of

$$\begin{cases} \mathbb{Z}/4\mathbb{Z}, & \text{if } k(\sqrt{d}) \subset k(\zeta_4) \\ \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2, & \text{if } k(\sqrt{d}) \subset k(\zeta_3), \\ \mathbb{Z}/2\mathbb{Z}, & \text{otherwise} \end{cases}$$

- (2) *Suppose that $H \cap X$ is an irreducible singular cubic curve C . If C is cuspidal, then*

$$\text{Br } U / \text{Br}_1 U \cong 0.$$

If C is nodal split multiplicative, then

$$\text{Br } U / \text{Br}_1 U \subseteq \mathbb{Z}/2.$$

If C is nodal non-split multiplicative where the corresponding quadratic extension is $L = k(\sqrt{d})$, $d \notin k^2$, then $\text{Br } U / \text{Br}_1 U$ is a subgroup of

$$\begin{cases} \mathbb{Z}/4\mathbb{Z}, & \text{if } k(\sqrt{d}) \subset k(\zeta_4) \\ \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2, & \text{if } k(\sqrt{d}) \subset k(\zeta_3) \\ \mathbb{Z}/2\mathbb{Z}, & \text{otherwise} \end{cases}$$

- (3) *Suppose that $C := H \cap X$ is geometrically the union of three lines and let L be the minimal extension over which they are defined. If the 3 lines intersect at an Eckard point, Then*

$$\text{Br } U / \text{Br}_1 U \cong 0.$$

If $L = k$ or L/k is a degree 3 Galois extension, then

$$\text{Br } U / \text{Br}_1 U \subset \mathbb{Z}/2\mathbb{Z}.$$

If $L \supset k(\sqrt{d})$ for some $d \notin k^2$, then $\text{Br } U / \text{Br}_1 U$ is a subgroup of

$$\begin{cases} \mathbb{Z}/4\mathbb{Z}, & \text{if } k(\sqrt{d}) \subset k(\zeta_4) \\ \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2, & \text{if } k(\sqrt{d}) \subset k(\zeta_3) \\ \mathbb{Z}/2\mathbb{Z}, & \text{otherwise.} \end{cases}$$

Proof. First, note that $\mathrm{Br} U / \mathrm{Br}_1 U \subset (\mathrm{Br} \bar{U})^{\Gamma_k}$. Since the Brauer group is a direct sum of its p -primary subgroups, it suffices to compute the Galois invariant subgroup of $\mathrm{Br} \bar{U}[n]$ for n is a power of p , for all primes p . The result now follows from Proposition 3.9, Proposition 3.10, Proposition 3.11, and Lemma Theorem 4.1. $\square$

Remark 4.3. In Corollary 4.2, the only possible subgroups of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ that may be realized by $\mathrm{Br}(\bar{U})^{\Gamma_k}$ are

$$\mathbb{Z}/2\mathbb{Z}, \text{ and } \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

Indeed, this follows by Lemma Theorem 4.1.

4.1. A class of examples. We construct a class of examples of affine cubic surfaces over $\mathbb{Q}$ where all the possibilities for $\mathrm{Br}(\bar{U})^{\Gamma_k}$ in Corollary 4.2(3) occur and $\mathrm{Br}(\bar{U})^{\Gamma_k}$ descends to $\mathrm{Br}(U)$.

Let k be a field of characteristic 0. The following class of smooth cubic surfaces is given by McKean in [12, §4.3]. Let $\{\alpha_1, \dots, \alpha_6\}$ be the roots of a degree 6 polynomial $H(t) \in k[t]$. Let $\{p_1, \dots, p_6\} \subset \mathbb{P}^2(\bar{k})$, where $p_i := [1 : \alpha_i : \alpha_i^3]$, for $i \in \{1, \dots, 6\}$. To ensure that $\{p_1, \dots, p_6\}$ are in general position, we require that $H(t) \in k[t]$ satisfies the following:

- (i) The roots of $H(t)$ are distinct.
- (ii) The degree 5 term in $H(t)$ is non-zero.
- (iii) No three roots of $H(t)$ sum to zero.

Since $H(t)$ is defined over k , the set of points $\{p_1, \dots, p_6\}$ is Γ_k -invariant. Therefore, the blow up $\pi : X \rightarrow \mathbb{P}_k^2$ of $\mathbb{P}_k^2$ at the set of points $\{p_1, \dots, p_6\}$ is a smooth cubic surface defined over k .

We use this construction to produce the following class of affine cubic surfaces. We note that the construction of the generator of $\mathrm{Br}(\bar{U})[n]$ in this example is inspired by [4, Theorem 4.2]. To descend $\mathrm{Br}(\bar{U})^{\Gamma_k}$ to k , we use a result of Demeio [6, Theorem 1.1] on Brauer groups of algebraic tori.

Proposition 4.4. *Let k be a number field. Let $F(t) \in k[t]$ be a degree 3 separable polynomial with roots $\{\alpha_1, \alpha_2, \alpha_3\}$ and Galois group G such that the coefficient of t^2 is non-zero. Let $\{\alpha_1, \dots, \alpha_6\}$ be the roots of the polynomial $F(t)F(t - a)$ for $a \in k^*$ ordered so that $\alpha_{i+3} = \alpha_i$ for $i \in \{1, 2, 3\}$. Let $\{p_1, \dots, p_6\} \subset \mathbb{P}^2(\bar{k})$, where $p_i := [1 : \alpha_i : \alpha_i^3]$. Then for almost all $a \in k^*$, the blow up $\pi : X \rightarrow \mathbb{P}_k^2$ of $\mathbb{P}_k^2$ at $\{p_1, \dots, p_6\}$ is a smooth cubic surface defined over k . Moreover, the following statements hold.*

- (1) *Let ℓ_i be the line joining the points p_i and p_{i+3} for $i = 1, 2, 3$. Then the action of G on the points p_i determines an action on the lines ℓ_i such that if $g \cdot p_i = p_{g(i)}$ then $g \cdot \ell_i = \ell_{g(i)}$ for $g \in G$.*
- (2) *Let U be the complement of the strict transform of $\cup_{i=1}^3 \ell_i$ in X . Then*

$$\mathrm{Br}(U) / \mathrm{Br}_1(U) \xrightarrow{\sim} \mathrm{Br}(\bar{U})^{\Gamma_k}.$$

- (3) *Let e_i be the exceptional curve over p_i . There exists $f, g \in \bar{k}(X)^*$ with divisors*

$$\begin{aligned} \mathrm{div}(f) &= [\ell_1] - [\ell_2] - [e_1] - [e_4] + [e_2] + [e_5] \\ \mathrm{div}(g) &= [\ell_2] - [\ell_3] - [e_2] - [e_5] + [e_3] + [e_6] \end{aligned} \tag{4.1}$$

such that the cyclic algebras

$$A_n := (f, g)_{\zeta_n} \in \text{Br } k(\bar{X})$$

generate $\text{Br}(\bar{U})[n]$ for all $n \in \mathbb{Z}_{\geq 0}$, for a fixed n -th root of unity $\zeta_n \in \bar{k}$.

Proof. First, we show that the polynomial $H(t) := F(t)F(t-a)$ satisfies the required conditions for the points to be in general position. Since $F(t)$ is separable, $H(t)$ has distinct roots for all but finitely many values of $a \in k$. Since the coefficient of t^2 in $F(t)$ is non-zero, no three roots of $H(t)$ will sum to zero for all but finitely many values of $a \in k$. Since the coefficient of the degree 5 term in $H(t)$ is the sum of the roots, it will be non-zero for all but finitely many values of $a \in k$. This shows that the blow up of $\mathbb{P}_k^2$ at $\{p_1, \dots, p_6\}$ is a smooth cubic surface X defined over k for almost all $a \in k$.

(1) follows straight away by the construction of the points $\{p_1, \dots, p_6\}$ and the lines $\{\ell_1, \ell_2, \ell_3\}$.

(2) Since $\pi : X \rightarrow \mathbb{P}_k^2$ is the blow up of $\mathbb{P}_k^2$ at the points $\{p_1, \dots, p_6\}$ which are contained in $\cup_{i=1}^3 \ell_i$, we deduce that

$$U \cong \mathbb{P}^2 \setminus \cup_{i=1}^3 \ell_i.$$

Let P be the singular locus of $\cup_{i=1}^3 \ell_i$. We may write $P = \text{Spec}(E)$ for some étale K -algebra E . If P is an Eckart point, then $\text{Br}(\bar{U}) = 0$ and the result holds. Thus, we may assume that E is an étale algebra of degree 3. Then, it follows that $\mathbb{P}^2 \setminus \cup_{i=1}^3 \ell_i$ is isomorphic to the torus

$$(\text{Res}_{E/k} \mathbb{G}_m) / \mathbb{G}_m.$$

Since k is a number field, we deduce by [6, Theorem 1.1] that the map $\text{Br}(U) \rightarrow \text{Br}(\bar{U})^{\Gamma_k}$ is surjective.

(3) First, we have $\text{Pic}(\bar{X}) = \mathbb{Z}[\ell] \oplus \bigoplus_{i=1}^6 [e_i]$ where ℓ is the inverse image of any line in $\mathbb{P}^2$ not containing any p_i . We can express the classes of the lines ℓ_i with respect to this basis as follows:

$$[\ell_1] = (3, 1, 0, 0, 1, 0, 0), [\ell_2] = (3, 0, 1, 0, 0, 1, 0), \text{ and } [\ell_3] = (3, 0, 0, 1, 0, 0, 1).$$

Therefore, we see that the divisors

$$\begin{aligned} D_1 &:= [\ell_1] - [\ell_2] - [e_1] - [e_4] + [e_2] + [e_5] \\ D_2 &:= [\ell_2] - [\ell_3] - [e_2] - [e_5] + [e_3] + [e_6] \end{aligned} \tag{4.2}$$

are principal. Thus, we have $D_1 = \text{div}(f)$ and $D_2 = \text{div}(g)$ for some $f, g \in \bar{k}(X)^*$.

Since $\cup_{i=1}^3 \ell_i$ intersects each e_i at a unique point, we have that $m_i := e_i \setminus (e_i \cap \cup_{i=1}^3 \ell_i)$ is isomorphic to $\mathbb{A}_k^1$. Let $V := U \setminus \cup_{i=1}^6 e_i$ which is defined over k as $\{e_1, \dots, e_6\}$ is Γ_k -invariant. Then by [3, Theorem 3.7.3], we have an exact sequence

$$0 \rightarrow \text{Br } \bar{U} \rightarrow \text{Br } \bar{V} \rightarrow \bigoplus_{i=1}^6 H_{\text{ét}}(m_i, \mathbb{Q}/\mathbb{Z}).$$

Since $m_i \cong \mathbb{A}_k^1$ and $H_{\text{ét}}(\mathbb{A}_k^1, \mathbb{Q}/\mathbb{Z}) = 0$, we obtain an isomorphism $\text{Br } \bar{U} \xrightarrow{\sim} \text{Br } \bar{V}$. For $n \in \mathbb{Z}_{>0}$ we fix an n -th root of unity $\zeta_n \in \bar{k}$. By our choice of f, g (see (4.2)), the cyclic algebra $A_n := (f, g)_{\zeta_n}$ is contained in $\text{Br}(\bar{V}) = \text{Br}(\bar{U})$. We claim that A_n

generates $(\text{Br } \bar{U})[n]$, which we know is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ as an abelian group (see Proposition 3.11). Indeed, by (4.2) we compute the residue of A_n at the generic point of ℓ_1 to be the class in $k(\ell_1)^\times/k(\ell_1)^{\times n}$ of the rational function induced by g on ℓ_1 which we denote by $g|_{\ell_1}$; the divisor of $g|_{\ell_1}$ is a linear combination of points in ℓ_1 , which contains the point $\ell_1 \cap \ell_3$ with multiplicity -1 . This shows that the order of the residue is n , i.e. A_n has order n and generates $(\text{Br } \bar{U})[n]$. $\square$

We use Proposition 4.4 to prove Theorem 1.2.

Proof of Theorem 1.2. We want to show that for all

$$A \in \{\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}\}$$

there exists an affine cubic surface U over $\mathbb{Q}$ that is the complement of three geometric lines such that $\text{Br}(U)/\text{Br}_1(U) \cong A$.

Let $F(t) \in \mathbb{Q}[t]$ be a degree 3 separable polynomial with Galois group $G = \text{Gal}(L/\mathbb{Q})$ such that the coefficient of t^2 is non-zero. If 2 divides $|G|$, then L contains a quadratic extension $\mathbb{Q}(\sqrt{d})$, for some $d \notin \mathbb{Q}^2$. We set $d = 1$ if $|G|$ is odd. Let U be an affine cubic surface given by Proposition 4.4 for some choice of $a \in k$ so that the lines do not meet an Eckard point. Then

$$\text{Br}(U)/\text{Br}_1(U) \xrightarrow{\sim} \text{Br}(\bar{U})^{\Gamma_k}.$$

By Proposition 3.11, we have that

$$\text{Br}(\bar{U}) \cong \begin{cases} \mathbb{Q}/\mathbb{Z}(-1), & \text{if } \text{Gal}(L/k) \cong \mathbb{Z}/3\mathbb{Z} \\ \varinjlim (M/nM(-1)), & \text{if } \text{Gal}(L/k) \cong S_3 \text{ or } \text{Gal}(L/k) \cong \mathbb{Z}/2\mathbb{Z} \end{cases}$$

By Lemma Theorem 4.1, we have

$$\varinjlim (M/nM(-1))^{\Gamma_k} \cong \begin{cases} \mathbb{Z}/2\mathbb{Z}, & \text{if } i \notin \mathbb{Q}(\sqrt{d}) \text{ and } \sqrt{-3} \notin \mathbb{Q}(\sqrt{d}), \\ \mathbb{Z}/4\mathbb{Z}, & \text{if } i \in \mathbb{Q}(\sqrt{d}), \\ \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}, & \text{if } \sqrt{-3} \in \mathbb{Q}(\sqrt{d}). \end{cases}$$

Therefore, to prove the result it suffices to find for each of the following Galois groups

- (1) $G = \text{Gal}(L/k)$ where $2 \mid |G|$ and $i, \sqrt{-3} \notin L$ (giving $\text{Br}(U)/\text{Br}_1(U) \cong \mathbb{Z}/2\mathbb{Z}$).
- (2) $G = \text{Gal}(L/k)$ where $2 \mid |G|$ and $i \in L$ (giving $\text{Br}(U)/\text{Br}_1(U) \cong \mathbb{Z}/4\mathbb{Z}$).
- (3) $G = \text{Gal}(L/k)$ where $2 \mid |G|$, $\sqrt{-3} \in L$ (giving $\text{Br}(U)/\text{Br}_1(U) \cong \mathbb{Z}/2 \times \mathbb{Z}/3$).

a degree 3 separable polynomial $F(t) \in \mathbb{Q}[t]$, such that the coefficient of t^2 is non-zero, with Galois group equal to G . Indeed, the following choices of $F(t)$ will give the desired Galois groups (respectively):

- (1) $F(t) = (t^2 - 2)(t + 1)$ has Galois group $\text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$.
- (2) $F(t) = (t^2 + 1)(t + 1)$ has Galois group $\text{Gal}(\mathbb{Q}(i)/\mathbb{Q})$.
- (3) $F(t) = (t^2 + 3)(t + 1)$ has Galois group $\text{Gal}(\mathbb{Q}(\sqrt{-3})/\mathbb{Q})$.

This completes the proof for all the possible groups A . $\square$

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